

Angular Interview Questions and Answers

What is Angular?

“Angular is a TypeScript-based front-end framework developed by Google used to build dynamic single-page web applications.”

Features of Angular

- Component-based architecture
 - Two-way data binding
 - Dependency Injection
 - Routing
 - TypeScript support
 - Reusable components
-

What is TypeScript?

“TypeScript is a superset of JavaScript that provides additional features like types, classes, interfaces, and compile-time checking.”

What is SPA?

“SPA stands for Single Page Application where the page loads once and updates content dynamically without full page refresh.”

Architecture

What is a Component?

“A component is the basic building block of Angular which controls a part of the UI.”

A component contains:

- HTML
- CSS
- TypeScript

What is a Module?

“A module is used to organize related components, directives, pipes, and services.”

Main module:

```
app.module.ts
```

What is Template?

“A template is an HTML view used to display UI in Angular.”

What is Decorator?

“Decorators are special functions used to provide metadata to classes.”

Examples:

- `@Component`
 - `@NgModule`
 - `@Injectable`
-

Data Binding

What is Data Binding?

“Data binding is communication between component and template.”

Types:

1. Interpolation
 2. Property Binding
 3. Event Binding
 4. Two-Way Binding
-

What is Interpolation?

“Interpolation displays TypeScript data in HTML using `{{ }}`.”

```
{{name}}
```

What is Property Binding?

“Property binding sends data from component to HTML.”

```
<img [src]="image">
```

What is Event Binding?

“Event binding handles user events like click and keypress.”

```
<button (click)="save()">
```

What is Two-Way Binding?

“Two-way binding is used for both displaying and updating data between component and view.”

```
<input [(ngModel)]="name">
```

Directives

What is Directive?

“Directives are used to modify DOM behavior.”

Types of Directives

1. Component Directive
 2. Structural Directive
 3. Attribute Directive
-

What is ngIf?

“*ngIf conditionally displays elements.”

```
<p *ngIf="show">Hello</p>
```

What is ngFor?

“*ngFor is used to iterate list data.”

```
<li *ngFor="let e of employees">
```

What is ngSwitch?

“ngSwitch is used for multiple conditional cases.”

What is ngClass?

“ngClass dynamically adds or removes CSS classes.”

What is ngStyle?

“ngStyle dynamically applies styles.”

Services & Dependency Injection

What is Service?

“A service is used to write reusable business logic.”

What is Dependency Injection?

“Dependency Injection is a design pattern where dependencies are provided automatically by Angular.”

What is Injectable?

“@Injectable marks a class as a service.”

Routing

What is Routing?

“Routing is navigation between components/pages.”

What is RouterOutlet?

“`router-outlet` is a placeholder where routed components are loaded.”

What is RouterLink?

“`routerLink` is used for navigation between pages.”

```
<a routerLink="/home">
```

Forms

Types of Forms in Angular

1. Template-Driven Forms
 2. Reactive Forms
-

What is Template-Driven Form?

“Form handled mainly in HTML using `ngModel`.”

What is Reactive Form?

“Reactive forms are handled in TypeScript using `FormControl` and `FormGroup`.”

Pipes

What is Pipe?

“Pipes transform displayed data.”

Example:

```
{{name | uppercase}}
```

Common Pipes

- uppercase
 - lowercase
 - date
 - currency
 - percent
-

HTTP & API

What is HttpClient?

“HttpClient is used to communicate with backend APIs.”

What is Observable?

“Observable handles asynchronous data like API responses.”

Difference between Promise and Observable?

- Promise returns single value
 - Observable returns multiple values
-

Lifecycle Hooks

What are Lifecycle Hooks?

“Lifecycle hooks are methods executed during component lifecycle.”

Common Lifecycle Hooks

- `ngOnInit()`
- `ngOnChanges()`
- `ngOnDestroy()`

What is `ngOnInit`?

“`ngOnInit()` executes after component initialization.”

Miscellaneous

What is Lazy Loading?

“Lazy loading loads modules only when required to improve performance.”

What is Eager Loading?

“Eager loading loads all modules at application startup.”

What is AOT Compilation?

“AOT converts Angular code into JavaScript during build time.”

What is DOM?

“DOM (Document Object Model) represents HTML elements as objects.”

What is Virtual DOM?

“Virtual DOM is a lightweight copy of the real DOM used for efficient updates.”

What is RxJS?

“RxJS is a library used for reactive programming with observables.”

What is Bootstrap in Angular?

“Bootstrap is used for styling and responsive UI design.”

Difference between AngularJS and Angular?

- AngularJS uses JavaScript
 - Angular uses TypeScript
 - Angular is faster and component-based
-

Advantages of Angular

- Reusable components
- Faster development
- Strong TypeScript support
- Good for SPA applications
- Easy API integration